Dokumentationsversion 01-00-00 Revision 2


```
1  {
2    "payload": {
3      "modelBaseVersionG": "3",
4      "profileNameG": "AFIR Energy Infrastructure",
5      "profileVersionG": "01-00-00",
6      "aegiEnergyInfrastructureTablePublication": {
7        "lang": "de",
8        "publicationTime": "2025-01-10T11:13:51+01:00",
9        "publicationCreator": {
10         "country": "DE",
11         "nationalIdentifier": "DE-NAP-OrganisationXY"
12       },
13       "headerInformation": {
14         "confidentiality": {"value": "noRestriction"},
15         "informationStatus": {"value": "test"}
16       },
17       "energyInfrastructureTable": [
18         {
19           "idG": "2474A514-0E5D-48F9-A908-F185DD4177A2",
20           "versionG": "2",
21           "tableName": "Region Nord",
22           "energyInfrastructureSite": [
23             {
24               "idG": "21F02723-CF84-4380-84D4-050917836C7C",
25               "versionG": "1",
26               "lastUpdated": "2025-01-10T11:13:51+01:00",
27               "accessibility": [
28                 {"value": "barrierFreeAccessible"}
```


Eine Table als reines Hierarchieelement

Ein Ladehub - Ansammlung von Ladestationen, auch in Verbindung mit Parkplätzen

```
29                  ],
30                  "additionalInformation": [
31                    {
32                      "values": [
33                        {
34                          "lang": "de",
35                          "value": "Zusatzinformationen"
36                        }
37                      ]
38                    }
39                  ],
40                  "informationWebsite": [
41                    {"urlLinkAddress": "https://---.com"}
42                  ],
43                  "operatingHours": {
44                    "afacOperatingHoursSpecification": {
45                      "idG": "1F38EC47-5BB7-47EE-B8E2-2680FA639FF1",
46                      "versionG": "1",
47                      "operatingAllYear": true,
48                      "overallPeriod": {
49                        "overallStartTime": "2025-01-01T00:00:00+01:00",
50                        "validPeriod": [
51                          {
52                            "recurringTimePeriodOfDay": [
53                              {
54                                "startTimeOfPeriod": "09:00:00Z",
55                                "endTimeOfPeriod": "17:00:00Z"
56                              }
```

Betriebszeiten müssen aus technischen Gründen mit einem nachgestellten "Z" angegeben werden - dies hat aber semantisch keine Bedeutung - es handelt sich hier **nicht** um eine UTC-Zeit!
Die Betriebszeiten hier sind 9 bis 17 Uhr.

```
57                          ]
58                        }
59                      ]
60                    }
61                  }
62                },
63                "locationReference": {
64                  "locAreaLocation": {
65                    "coordinatesForDisplay": {
66                      "latitude": 50.779599,
67                      "longitude": 6.104507
68                    },
69                    "locLocationExtensionG": {
70                      "FacilityLocation": {
71                        "address": {
72                          "postcode": "55555",
73                          "city": {
74                            "values": [
75                              {
76                                "lang": "de",
77                                "value": "Musterstadt"
78                              }
79                            ]
80                          },
81                          "countryCode": "DE",
82                          "addressLine": [
83                            {
84                              "order": 0,
```

Georeferenzierung der Site - Optional, aber empfohlen.

Möglichkeiten: WGS84-Koordinaten, OpenLR Punkt oder Fläche, GML Polygon, Adresse.

Wird eine Flächendefinition verwendet, können Punkt-Koordinaten über coordinatesForDisplay mitgegeben werden.

Falls kein geeigneterer Punkt bekannt ist, können die Koordinaten der ersten Ladestation verwendet werden.

Ein- und Ausfahrten können bei Bedarf an anderer Stelle eigenständig spezifiziert werden

Die Großschreibung ist an dieser Stelle korrekt und hat mit dem hier genutzten DATEX II Erweiterungsmechanismus zu tun

```
85                                  "type": {"value": "street"},
86                                  "text": {
87                                    "values": [
88                                      {
89                                        "lang": "de",
90                                        "value": "Hauptstraße"
91                                      }
92                                    ]
93                                  }
94                                },
95                                {
96                                  "order": 1,
97                                  "type": {"value": "houseNumber"},
98                                  "text": {
99                                    "values": [
100                                     {
101                                       "lang": "de",
102                                       "value": "12"
103                                     }
104                                   ]
105                                 }
106                               }
107                             ]
108                           }
109                         }
110                       },
111                       "gmlMultiPolygon": {
112                         "gmlPolygon": [
```

```
113                         {
114                           "exterior": {
115                             "srsDimension": 2,
116                             "srsName": "EPSG 4326 WGS84",
117                             "posList": "50.77978 6.10437 50.7797 6.10411 50.7795 6.10422
117 50.77956 6.10473 50.77982 6.10469 50.77978 6.10437"
118                           }
119                         }
120                       ]
121                     }
122                   }
123                 },
124                 "operator": {
125                   "afacAnOrganisation": {
126                     "name": {
127                       "values": [
128                         {
129                           "lang": "de",
130                           "value": "Operator XYZ"
131                         }
132                       ]
133                     },
134                     "legalName": {
135                       "values": [
136                         {
137                           "lang": "de",
138                           "value": "Operator XYZ"
139                         }
```

Das Polygon sollte gegen den Uhrzeigersinn gebildet werden und geschlossen sein (gleiche Anfangs- und Endkoordinaten)

Hier auch OpenLR möglich

Die Angabe eines Betreibers mit Operator ID ist verpflichtend (siehe weiter unten), außerdem auch Commercial- und legal name

Commercial name; zu beachten: Die EU-Regulierung verlangt sowohl den Commercial name als auch den Legal name des Betreibers (nachfolgendes Element)

```
140                     ]
141                   },
142                   "organisationUnit": [
143                     {
144                       "name": {
145                         "values": [
146                           {
147                             "lang": "de",
148                             "value": "Niederlassung West"
149                           }
150                         ]
151                       },
152                       "contactInformation": [
153                         {
154                           "afacContactInformation": {"eMail": "email@operator.com"}
155                         }
156                       ]
157                     }
158                   ],
159                   "externalIdentifier": [
160                     {
161                       "identifier": "DE*XYZ",
162                       "typeOfIdentifier": {
163                         "value": "extendedG",
164                         "extendedValueG": "operatorId"
165                       }
166                     },
167                     {
```

```
168                     "identifier": "[BNetzA-ID des Operators]",
169                     "typeOfIdentifier": {
170                       "value": "extendedG",
171                       "extendedValueG": "operatorIdBNetzA"
172                     }
173                   }
174                 ]
175               }
176             },
177             "dedicatedParkingSpaces": [
178               {
179                 "idG": "13D7BE2D-07BC-4
180                 "versionG": "1",
181                 "applicableForVehicles"
182                   {
183                     "vehicleType": [
184                       {"value": "car"}
185                     ]
186                   }
187                 ],
188                 "numberOfSpaces": 25
189               },
190               {
191                 "idG": "0F6682EC-0884-499B-AE29-8DFFF510945F",
192                 "versionG": "1",
193                 "applicableForVehicles": [
194                   {
195                     "vehicleType": [
```

Option

Auf Site-Ebene, optional: Anzahl der Park

```
196                         {"value": "lorry"}
197                       ]
198                     }
199                   ],
200                   "numberOfSpaces": 0
201                 }
202               ],
203               "typeOfSite": {"value": "openSpace"},
204               "brand": {
205                 "values": [
206                   {
207                     "lang": "de",
208                     "value": "E-Energy"
209                   }
210                 ]
211               },
212               "exclusiveUsers": [
213                 {"value": "allUsers"}
214               ],
215               "serviceType": [
216                 {
217                   "serviceType": {"value": "physicalAttendance"},
218                   "overallPeriod": {
219                     "overallStartTime": "2024-01-01T00:00:00+01:00",
220                     "overallEndTime": "2026-12-31T23:59:00+01:00",
221                     "validPeriod": [
222                       {
223                         "recurringTimePeriodOfDay": [
```

Hier w

```
224                         {
225                           "startTimeOfPeriod": "07:00:00Z",
226                           "endTimeOfPeriod": "20:00:00Z"
227                         }
228                       ],
229                       "recurringDayWeekMonthPeriod": [
230                         {
231                           "comDayWeekMonth": {
232                             "applicableDay": [
233                               {"value": "monday"},
234                               {"value": "tuesday"},
235                               {"value": "wednesday"},
236                               {"value": "thursday"},
237                               {"value": "friday"}
238                             ]
239                           }
240                         }
241                       ]
242                     }
243                   ]
244                 }
245               }
246             ],
247             "entrance": [
248               {
249                 "locPointLocation": {
250                   "coordinatesForDisplay": {
251                     "latitude": 50.77959,
```

Physikalische Anwesenheit von Personal MO-FR von 7 bis 20 Uhr

Optional: Zufahrt zur Site (es können mehrere Zu- und Abfahrten angegeben werden)

Zunächst Koordinaten für eine Map-Ansicht (hier im Beispiel unverändert zur Site, sonst aber sinnvollerweise nicht exakt identisch)

```
252                     "longitude": 6.1045
253                   },
254                   "supplementaryPositionalDescription": {
255                     "locationDescription": {
256                       "values": [
257                         {
258                           "lang": "de",
259                           "value": "Zusatzinformationen zur Einfahrt"
260                         }
261                       ]
262                     }
263                   },
264                   "pointByCoordinates": {
265                     "pointCoordinates": {
266                       "latitude": 50.77959
267                       "longitude": 6.1045
268                     }
269                   }
270                 }
271               }
272             ],
273             "mobilityServiceProvider": [
274               {
275                 "afacAnOrganisation": {
276                   "name": {
277                     "values": [
278                       {
279                         "lang": "de",
```

Optionale Z

WGS84 K

Vertragspa

```
280                         "value": "Mobility service provider XY"
281                       }
282                     ]
283                   },
284                   "available24hours": false,
285                   "legalName": {
286                     "values": [
287                       {
288                         "lang": "de",
289                         "value": "Mobility service provider XY GmbH"
290                       }
291                     ]
292                   },
293                   "linkToLogo": "https://___________",
294                   "type": {"value": "privateSector"},
295                   "organisationUnit": [
296                     {
297                       "contactInformation": [
298                         {
299                           "afacContactPerson": {
300                             "name": "Max Mustermann",
301                             "eMail": "max.mustermann@mail.de"
302                           }
303                         }
304                       ]
305                     }
306                   ],
307                   "externalIdentifier": [
```

```
308                     {
309                       "identifier": "DE*XYZ",
310                       "typeOfIdentifier": {
311                         "value": "extendedG",
312                         "extendedValueG": "providerId"
313                       }
314                     }
315                   ]
316                 }
317               }
318             ],
319             "energyInfrastructureStation": [
320               {
321                 "idG": "68722A13-ECD6-4A51-8D6D-01A933F2D3DF",
322                 "versionG": "1",
323                 "lastUpdated": "2025-01-10T11:13:51+01:00",
324                 "description": {
325                   "values": [
326                     {
327                       "lang": "de",
328                       "value": "evt. genauere Beschreibung"
329                     }
330                   ]
331                 },
332                 "accessibility": [
333                   {"value": "barrierFreeAccessible"}
334                 ],
335                 "additionalInformation": [
```

Falls ei

Erste La

```
336                     {
337                       "values": [
338                         {
339                           "lang": "de",
340                           "value": "evt. Zusatzinformationen"
341                         }
342                       ]
343                     }
344                   ],
345                   "externalIdentifier": [
346                     {
347                       "identifier": "[Ladestations-ID der Bundesnetzagentur]",
348                       "typeOfIdentifier": {
349                         "value": "extendedG",
350                         "extendedValueG": "stationIdBNetzA"
351                       }
352                     }
353                   ],
354                   "informationWebsite": [
355                     {"urlLinkAddress": "https://--------"}
356                   ],
357                   "photoUrl": [
358                     {"urlLinkAddress": "https://--------"}
359                   ],
360                   "operatingHours": {
361                     "afacOpenAllHours": {}
362                   },
363                   "locationReference": {
```

Optiona

```
364                 "locPointLocation": {
365                   "coordinatesForDisplay":
366                     "latitude": 50.77959,
367                     "longitude": 6.1045
368                   },
369                   "locLocationExtensionG": {
370                     "FacilityLocation": {
371                       "timeZone": "+01:00",
372                       "address": {
373                         "postcode": "55555",
374                         "city": {
375                           "values": [
376                             {
377                               "lang":
378                               "value":
379                             }
380                           ]
381                         },
382                         "countryCode": "DE",
383                         "addressLine": [
384                           {
385                             "order": 0,
386                             "type": {"value": "street"},
387                             "text": {
388                               "values": [
389                                 {
390                                   "lang": "de",
391                                   "value": "Hauptstraße"
```

Optional: W

Für die Geo

Verpflichtende

```
392                                 }
393                               ]
394                             }
395                           },
396                           {
397                             "order": 1,
398                             "type": {"value": "houseNumber"},
399                             "text": {
400                               "values": [
401                                 {
402                                   "lang": "de",
403                                   "value": "12"
404                                 }
405                               ]
406                             }
407                           }
408                         ]
409                       },
410                       "nutsArea": [
411                         {
412                           "nutsCodeType": {"value": "nuts1Code"},
413                           "nutsCode": "DEA"
414                         }
415                       ]
416                     }
417                   },
418                   "supplementaryPositionalDescription": {
419                     "locationDescription": {
```

Verpflichtende Angabe des NUTS-Codes für den Standort (Bundesland) der Ladestation.

Hier 'DEA' für Nordrhein-Westfalen

(NUTS-Codes weiterer Ebenen optional)

```
420                         "values": [
421                           {
422                             "lang": "de",
423                             "value": "Zusatzinformationen"
424                           }
425                         ]
426                       }
427                     },
428                     "pointByCoordinates": {
429                       "pointCoordinates": {
430                         "latitude": 50.77959,
431                         "longitude": 6.1045
432                       }
433                     }
434                   }
435                 },
436                 "operator": {
437                   "afacAnOrganisation": {
438                     "name": {
439                       "values": [
440                         {
441                           "lang": "de",
442                           "value": "[Name des Betre
443                         }
444                       ]
445                     },
446                     "legalName": {
447                       "values": [
```

Zusatzinformationen (Freitext), z.B. auch über Parkdeck etc.; außerdem Präzisionsangabe f.d. Georeferenzierung möglich

Verpflichtende Koordinaten-Angaben WGS84

Hier auch OpenLR möglich

Verpflic

```
448                         {
449                           "lang": "de",
450                           "value": "[Offizielle Bezeichnung des Betreibers]"
451                         }
452                       ]
453                     },
454                     "externalIdentifier": [
455                       {
456                         "identifier": "DE*XYZ",
457                         "typeOfIdentifier": {
458                           "value": "extendedG",
459                           "extendedValueG": "op
460                         }
461                       },
462                       {
463                         "identifier": "[BNetzA-ID des Operators]",
464                         "typeOfIdentifier": {
465                           "value": "extendedG",
466                           "extendedValueG": "operatorIdBNetzA"
467                         }
468                       }
469                     ]
470                   }
471                 },
472                 "helpdesk": {
473                   "afacReferenc
474                     "idG": "DC
475                     "versionG": "2",
```

Verpflic

Hier viele w

verpflichtende

Option

verpflichtende Ar

```
476                         "name": {
477                           "values": [
478                             {
479                               "lang": "de",
480                               "value": "Ihre Helpdesk GmbH"
481                             }
482                           ]
483                         },
484                         "organisationUnit": [
485                           {
486                             "contactInformation": [
487                               {
488                                 "afacContactInformation": {"telephoneNumber":
488 "+4924100000000"}
489                               }
490                             ],
491                             "operatingHours": {
492                               "afacOperatingHoursSpecification": {
493                                 "idG": "D3AD3210-CCD0-4AD5-9E70-B582251F69AA",
494                                 "versionG": "3",
495                                 "overallPeriod": {
496                                   "overallStartTime": "2025-01-01T00:00:00+01:00",
497                                   "overallEndTime": "2025-12-31T23:59:00+01:00",
498                                   "validPeriod": [
499                                     {
500                                       "recurringTimePeriodOfDay": [
501                                         {
502                                           "startTimeOfPeriod": "06:00:00Z",
```

verpflichtende

Verfügbarkeit des Helpde

```
503                                         "endTimeOfPeriod": "23:00:00Z"
504                                       }
505                                     ]
506                                   }
507                                 ],
508                                 "exceptionPeriod": [
509                                   {
510                                     "recurringDayWeekMonthPeriod": [
511                                       {
512                                         "comDayWeekMonth": {
513                                           "applicableDay": [
514                                             {"value": "sunday"}
515                                           ]
516                                         }
517                                       }
518                                     ],
519                                     "recurringSpecialDay": [
520                                       {
521                                         "comSpecialDay": {
522                                           "intersectWithApplicableDays": false,
523                                           "specialDayType": {"value": "publicHoliday"}
524                                         }
525                                       }
526                                     ]
527                                   },
528                                   {
529                                     "startOfPeriod": "2025-07-15T00:00:00+02:00",
530                                     "endOfPeriod": "2025-07-22T23:59:59+02:00"
```

```
531                                 }
532                               ]
533                             }
534                           }
535                         }
536                       }
537                     ]
538                   }
539                 },
540                 "applicableForVehicles": [
541                   {
542                     "grossWeightCharacteristic": [
543                       {
544                         "comparisonOperator": {"value": "lessThan"},
545                         "grossVehicleWeight": 2,
546                         "typeOfWeight": {"value": "actual"}
547                       }
548                     ],
549                     "comVehicleCharacteristicsExtensionG": {
550                       "euVehicleCategory": [
551                         {"value": "l"},
552                         {"value": "m1"}
553                       ]
554                     }
555                   }
556                 ],
557                 "amenities": {
558                   "illuminated": false,
```

Erlaubt sind

Überdacht a

```
559                     "roofed": true,
560                     "illuminatedRechargingParkingNearby": true
561                   },
562                   "supplementalFacility": [
563                     {
564                       "afacSupplementalEquipment": {
565                         "idG": "AF09B7AC-B5B1-4FA1-AACF-E
566                         "versionG": "1",
567                         "quantity": 2,
568                         "equipmentType": {"value": "paymentMachine"}
569                       }
570                     }
571                   ],
572                   "dedicatedParkingSpaces": [
573                     {
574                       "idG": "A6450D0F-1B74-4B8
575                       "versionG": "1",
576                       "accessibility": [
577                         {"value": "barrierFreeAccessible"}
578                       ],
579                       "applicableForVehicles": [
580                         {
581                           "comVehicleCharacteristicsExtensionG": {
582                             "euVehicleCategory": [
583                               {"value": "l"},
584                               {"value": "m1"}
585                             ]
586                           }
```

Die EU-Reg

Unmittelbar zu

```
587                         }
588                       ],
589                       "amenities": {
590                         "illuminated": false,
591                         "roofed": true
592                       },
593                       "numberOfSpaces"
594                     },
595                     {
596                       "idG": "A6450D0F-1B74-4B8E-9388-E7A252FC0B66",
597                       "versionG": "1",
598                       "locationReference": {
599                         "locPointLocation": {
600                           "pointByCoordinates": {
601                             "pointCoordinates": {
602                               "latitude": 50.779599,
603                               "longitude": 6.104507
604                             }
605                           }
606                         }
607                       },
608                       "dimension": {"width": 4},
609                       "numberOfSpaces": 1,
610                       "userSpecific": [
611                         {"value": "personsWithDisabilities"}
612                       ]
613                     }
614                   ],
```

Ein weiterer Plat

Koo

```
615                 "totalMaximumPower": 100000,
616                 "authenticationAndIdentificationMethods": [
617                   {"value": "activeRFIDChip"},
618                   {"value": "creditCard"}
619                 ],
620                 "numberOfRefillPoints": 2,
621                 "userInterfaceLanguage": ["
622                 "energyDistributor": {
623                   "afacOrganisationByReference": {
624                     "organisationReference": {
625                       "targetClass": "afacIdentifiableOrganisation",
626                       "idG": "01B5CCA0-E746-4084-9EEB-47F0DB297AB0",
627                       "versionG": "1"
628                     }
629                   }
630                 },
631                 "mobilityServiceProvider": [
632                   {
633                     "afacAnOrganisation": {
634                       "name": {
635                         "values": [
636                           {
637                             "lang": "de",
638                             "value": "Stromanbieter ABC3"
639                           }
640                         ]
641                       },
642                       "externalIdentifier": [
```

Die Ladestat

Netzbetre

```
643                         {
644                           "identifier": "DE*XYZ",
645                           "typeOfIdentifier": {
646                             "value": "extendedG",
647                             "extendedValueG": "providerId"
648                           }
649                         }
650                       ]
651                     }
652                   }
653                 ],
654                 "roamingPlatform": [
655                   {
656                     "afacAnOrganisation": {
657                       "name": {
658                         "values": [
659                           {
660                             "lang": "de",
661                             "value": "Roaming1"
662                           }
663                         ]
664                       }
665                     }
666                   },
667                   {
668                     "afacAnOrganisation": {
669                       "name": {
670                         "values": [
```

```
671                                 {
672                                   "lang": "de",
673                                   "value": "Roaming2"
674                                 }
675                               ]
676                             }
677                           }
678                         },
679                         {
680                           "afacAnOrganisation": {
681                             "name": {
682                               "values": [
683                                 {
684                                   "lang": "de",
685                                   "value": "Roaming3"
686                                 }
687                               ]
688                             }
689                           }
690                         }
691                       ],
692                       "serviceType": [
693                         {
694                           "serviceType": {"value": "unattended"}
695                         }
696                       ],
697                       "refillPoint": [
698                         {
```

1. La

```
699                     "aegiElectricChargingPoint": {
700                       "idG": "73ABE928-707D-4A99-8043-4293EE685504",
701                       "versionG": "2",
702                       "lastUpdated": "2025-01-10T11:13:51+01:00",
703                       "externalIdentifier": [
704                         {
705                           "identifier": "DE*ABC*EXXXXXXXX12345",
706                           "typeOfIdentifier": {
707                             "value": "extendedG",
708                             "extendedValueG": "evseI
709                           }
710                         }
711                       ],
712                       "deliveryUnit": {"value": "kWh"},
713                       "modelType": {
714                         "values": [
715                           {
716                             "lang": "de",
717                             "value": "Anschlusstyp AB123"
718                           }
719                         ]
720                       },
721                       "currentType": {"value": "ac"},
722                       "usageType": [
723                         {"value": "electricVehicle"}
724                       ],
725                       "vehicleToGridCommunicationType": [
726                         {"value": "iso15118"}
```

Die EVS-ID nac

```
727                         ],
728                         "numberOfConnectors": 2,
729                         "availableVoltage": [
730                           200,
731                           400
732                         ],
733                         "availableChargingPower": [
734                           50000,
735                           22000,
736                           11000
737                         ],
738                         "smartRechargingServices": [
739                           {"value": "remoteMonitoring"},
740                           {"value": "bidirectionalRecharging"},
741                           {"value": "powerOptimisationByUser"},
742                           {"value": "plugAndCharge"}
743                         ],
744                         "otherSmartRechargingServices": [
745                           {
746                             "values": [
747                               {
748                                 "lang": "de",
749                                 "value": "weitere smart recharging Services"
750                               }
751                             ]
752                           }
753                         ],
754                         "connector": [
```

2 vers

Mögliche

Angabe

```
755                         {
756                           "connectorType": {"value": "iec62196T2"},
757                           "maxPowerAtSocket": 22000
758                         },
759                         {
760                           "connectorType": {"value": "chademo"},
761                           "maxPowerAtSocket": 50000
762                         }
763                       ],
764                       "electricEnergy": [
765                         {
766                           "energyProductName": {
767                             "values": [
768                               {
769                                 "lang": "de"
770                                 "value": "1a
771                               }
772                             ]
773                           },
774                           "energyProductInformation": "h
775                           "renewableEnergyEvidence": "https://                    ",
776                           "energyRate": [
777                             {
778                               "idG": "74034E3E-9D2F-4410-BE6F-CAA
779                               "ratePolicy": {"value": "adHoc"},
780                               "lastUpdated": "2025-01-10T11:13:51
781                               "applicableCurrency": ["EUR"],
782                               "payment": {
```

Eine (nicht näher spezifiz

Zur Förder

Webseite

Link direkt auf ei

Preisgestaltu

Achtung: Eine AdHoc-P

Ad-H

Eine weite

```
783                                 "paymentMeans": [
784                                   {"value": "emv"},
785                                   {"value": "website"}
786                                 ],
787                                 "brandsAccepted": [
788                                   {"value": "visa"}
789                                 ]
790                               },
791                               "energyPrice": [
792                                 {
793                                   "priceType": {"value": "pricePerKWh"},
794                                   "value": 0.35
795                                 }
796                               ]
797                             }
798                           ]
799                         },
800                         {
801                           "energyProductName": {
802                             "values": [
803                               {
804                                 "lang": "de",
805                                 "value": "Grüner Strom"
806                               }
807                             ]
808                           },
809                           "mobilityServiceProvider": [
810                             {
```

Referenz auf den zuv

Nachtp

```
811                 "afacOrganisationByReference": {
812                   "organisationReference": {
813                     "targetClass": "afacIdentifiableOrganisation",
814                     "idG": "A1633959-F0BE-4F48-85CC-1A9FAE2BF34B"
815                   }
816                 }
817
818               ],
819               "energyRate": [
820                 {
821                   "idG": "9C07640B-7ECE-46EC-A23F-417E99C53F33",
822                   "ratePolicy": {"value": "contract"},
823                   "lastUpdated": "2025-01-10T11:13:51+01:00",
824                   "applicableCurrency": ["EUR"],
825                   "rateName": {
826                     "values": [
827                       {
828                         "lang": "de",
829                         "value": "GrünTarif 1AB"
830                       }
831                     ]
832                   },
833                   "energyPrice": [
834                     {
835                       "priceType": {"value": "pricePerKWh"},
836                       "value": 0.3,
837                       "overallPeriod": {
838                         "overallStartTime": "2025-01-01T00:00:00+01:00",
```

```
839                                         "overallEndTime": "2025-12-31T23:59:59+01:00",
840                                         "validPeriod": [
841                                           {
842                                             "recurringTimePeriodOfDay": [
843                                               {
844                                                 "startTimeOfPeriod": "21:00:00Z",
845                                                 "endTimeOfPeriod": "06:00:00Z"
846                                               }
847                                             ]
848                                           }
849                                         ]
850                                       }
851                                     }
852                                   ]
853                                 }
854                               ]
855                             }
856                           ]
857                         }
858                       },
859                       {
860                         "aegiElectricChargingPoint": {
861                           "idG": "35E5FC89-E2C9-4946-8E70-6A2C56E0BD7E",
862                           "versionG": "1",
863                           "lastUpdated": "2025-01-10T11:13:51+01:00",
864                           "deliveryUnit": {"value": "kWh"},
865                           "currentType": {"value": "ac"},
866                           "usageType": [
```

2. Ladepunkt

```
867                         {"value": "electricVehicle"}
868                       ],
869                       "vehicleToGridCommunicationType": [
870                         {"value": "iec619802"}
871                       ],
872                       "numberOfConnectors": 1,
873                       "availableVoltage": [400],
874                       "availableChargingPower": [40000],
875                       "smartRechargingServices": [
876                         {"value": "remoteMonitoring"}
877                       ],
878                       "connector": [
879                         {
880                           "connectorType": {"value": "other"},
881                           "otherConnector": "Some other connector type",
882                           "maxPowerAtSocket": 40000
883                         }
884                       ],
885                       "electricEnergy": [
886
887                           "energyRateByReference": [
888                             {
889                               "targetClass": "aegiEnergyRate",
890                               "idG": "9C07640B-7ECE-46EC-A23F-417E99C53F33"
891                             }
892                           ]
893                         }
894                       ]
```

Der 2. Ladepunkt nutz

```
895                     }
896                   }
897                 ],
898                 "electricEnergy": [
899                   {
900                     "isGreenEnergy": true,
901                     "mobilityServiceProvider": [
902                       {
903                         "afacOrganisationByReference": {
904                           "organisationReference": {
905                             "targetClass": "afacIdentifiableOrganisation",
906                             "idG": "A1633959-F0BE-4F48-85CC-1A9FAE2BF34B"
907                           }
908                         }
909                       }
910
911                     energyProductInformation": "https://_____
912                     renewableEnergyEvidence": "https://_____ ",
913                     energyRate": [
914                       {
915                         "idG": "A1633959-F0BE-4F48-85CC-1A9FAE2BF34B",
916                         "ratePolicy": {"value": "adHoc"},
917                         "lastUpdated": "2025-01-10T11:13:51+01:00",
918                         "applicableCurrency": ["EUR"],
919                         "payment": {
920                           "paymentMeans": [
921                             {"value": "emv"},
922                             {"value": "nfc"}
```

Auf Ladestations-Ebene: V

Hinweise sie

Hinweis: Diese Bezahlmethoden implizieren im Sinne der EU-Regulierung, dass ein entsprechender Bezahl-Automat (vermutlich integriert in die Ladestation) vorhanden ist

```
923                                 ]
924                               }
925                             }
926                           ]
927                         },
928                         {
929                           "isGreenEnergy": true,
930                           "mobilityServiceProvider": [
931                             {
932                               "afacReferenceableOrganisation": {
933                                 "idG": "A1633959-F0BE-4F48-85CC-1A9FAE2BF34B",
934                                 "versionG": "1",
935                                 "name": {
936                                   "values": [
937                                     {
938                                       "lang": "de",
939                                       "value": "Vertragsparter XY"
940                                     }
941                                   ]
942                                 },
943                                 "linkToGeneralInformation": "linkToGeneralInformation",
944                                 "linkToWebform": "https://contact.me",
945                                 "externalIdentifier": [
946                                   {
947                                     "identifier": "DE*XYZ",
948                                     "typeOfIdentifier": {
949                                       "value": "extendedG",
950                                       "extendedValueG": "providerId"
```

Der Vertragspartner wird später nochmals referenziert, daher als Referenceable Organisation

```
                                    }
                                  }
                                ]
                              }
                            }
                          ],
                          "energyRate": [
                            {
                              "idG": "4C55D08A-20D5-4A7A-8067-81AB5E15E738",
                              "ratePolicy": {"value": "contract"},
                              "lastUpdated": "2025-01-10T11:13:51+01:00",
                              "applicableCurrency": ["EUR"],
                              "payment": {
                                "paymentMeans": [
                                  {"value": "qrCode"},
                                  {"value": "website"}
                                ]
                              }
                            }
                          ]
                        }
                      ]
                    },
                    {
                      "idG": "0563BFAD-646D-4A19-9E5C-6D4599FAAF6A",
                      "versionG": "3",
                      "lastUpdated": "2025-01-10T11:13:51+01:00",
                      "accessibility": [
```


Vertrags-Str

Zweite L

ACHTUNG: Das Beispiel der 2. Ladestation ist nicht mehr vollständig ausgearbeitet. Es ist syntaktiscjh korrekt, enthält aber nicht alle verpflichtenden Elemente, sondern nurmehr eine Anschauung weiterer Möglichkeiten

```
979                 {"value": "disabilityAccessible"}
980               ],
981               "externalIdentifier": [
982                 {
983                   "identifier": "[Ladestations-ID der Bundesnetzagentur]",
984                   "typeOfIdentifier": {
985                     "value": "extendedG",
986                     "extendedValueG": "stationIdBNetzA"
987                   }
988                 }
989               ],
990               "locationReference": {
991                 "locPointLocation": {
992                   "coordinatesForDisplay": {
993                     "latitude": 50.779594,
994                     "longitude": 50.779594
995                   }
996                 }
997               },
998               "dedicatedParkingSpaces": [
999                 {
1000                  "idG": "BC3BBA0E-14B1-461E-83B2-1CF5A011
1001                  "versionG": "1",
1002                  "accessibility": [
1003                    {"value": "barrierFreeAccessible"}
1004                  ],
1005                  "locationReference": {
1006                    "locAreaLocation": {
```


Optionale

Exakte Punkt-Koord

Unmittelbar zur Lade

```
1007                            "gmlMultiPolygon": {
1008                              "gmlPolygon": [
1009                                {
1010                                  "exterior": {
1011                                    "srsDimension": 2,
1012                                    "srsName": "EPSG 4326 WGS84",
1013                                    "posList": "50.77967 6.10438 50.77964 6.10439
1013  50.77964 6.10448 50.77969 6.10447 50.77977 6.10445 50.77967 6.10438"
1014                                  }
1015                                }
1016                              ]
1017                            }
1018                          }
1019                        },
1020                        "applicableForVehicles": [
1021                          {
1022                            "fuelType": [
1023                              {"value": "battery"}
1024                            ],
1025                            "vehicleType": [
1026                              {"value": "car"}
1027                            ],
1028                            "heightCharacteristic": [
1029                              {
1030                                "comparisonOperator": {"value": "lessThan"},
1031                                "vehicleHeight": 3
1032                              }
1033                            ]
```

Beschreibung d

Das Polygon sollte

Stellpla

```
1034                    }
1035                  ],
1036                  "dimension": {
1037                    "length": 6,
1038                    "width": 3
1039                  },
1040                  "numberOfSpaces": 1
1041                },
1042                {
1043                  "idG": "BC44BBE7-DFD5-4822-BF51-30D213A4CF9F",
1044                  "versionG": "1",
1045                  "numberOfSpaces": 0,
1046                  "userSpecific": [
1047                    {"value": "personsWithDisabilities"}
1048                  ]
1049                }
1050              ],
1051              "totalMaximumPower": 200000,
1052              "authenticationAndIdentificationMethods": [
1053                {"value": "apps"}
1054              ],
1055              "numberOfRefillPoints": 2,
1056              "userInterfaceLanguage": ["de"],
1057              "mobilityServiceProvider": [
1058                {
1059                  "afacAnOrganisation": {
1060                    "name": {
1061                      "values": [
```

```
1062                                {
1063                                  "lang": "de",
1064                                  "value": "Stromanbieter ABC3"
1065                                }
1066                              ]
1067                            },
1068                            "externalIdentifier": [
1069                              {
1070                                "identifier": "DE*XYZ",
1071                                "typeOfIdentifier": {
1072                                  "value": "extendedG",
1073                                  "extendedValueG": "providerId"
1074                                }
1075                              }
1076                            ]
1077                          }
1078                        }
1079                      ],
1080                      "roamingPlatform": [
1081                        {
1082                          "afacAnOrganisation": {
1083                            "name": {
1084                              "values": [
1085                                {
1086                                  "lang": "de",
1087                                  "value": "RPT1"
1088                                }
1089                              ]
```

```
1090                        }
1091                      }
1092                    },
1093                    {
1094                      "afacAnOrganisation": {
1095                        "name": {
1096                          "values": [
1097                            {
1098                              "lang": "de",
1099                              "value": "RPT2"
1100                            }
1101                          ]
1102                        }
1103                      }
1104                    }
1105                  ],
1106                  "serviceType": [
1107                    {
1108                      "serviceType": {"value": "unattended"}
1109                    }
1110                  ],
1111                  "refillPoint": [
1112                    {
1113                      "aegiElectricChargingPoint": {
1114                        "idG": "CAEBDA8A-210A-48EA-856A-EA9595FDDD10",
1115                        "versionG": "2",
1116                        "lastUpdated": "2025-01-10T11:13:51+01:00",
1117                        "deliveryUnit": {"value": "kWh"},
```

1. Ladepunkt

```
1118                    "modelType": {
1119                      "values": [
1120                        {
1121                          "lang": "de",
1122                          "value": "Anschlusstyp XY"
1123                        }
1124                      ]
1125                    },
1126                    "currentType": {"value": "ac"},
1127                    "usageType": [
1128                      {"value": "electricVehicle"}
1129                    ],
1130                    "vehicleToGridCommunicationType": [
1131                      {"value": "iso15118"}
1132                    ],
1133                    "numberOfConnectors": 2,
1134                    "availableVoltage": [
1135                      200,
1136                      400
1137                    ],
1138                    "availableChargingPower": [
1139                      50000,
1140                      22000,
1141                      11000
1142                    ],
1143                    "smartRechargingServices": [
1144                      {"value": "remoteMonitoring"},
1145                      {"value": "bidirectionalRecharging"},
```

2 versc

```
1146                    {"value": "powerOptimisationByUser"},
1147                    {"value": "plugAndCharge"}
1148                  ],
1149                  "otherSmartRechargingServices": [
1150                    {
1151                      "values": [
1152                        {
1153                          "lang": "de",
1154                          "value": "weitere smart recharging Services"
1155                        }
1156                      ]
1157                    }
1158                  ],
1159                  "connector": [
1160                    {
1161                      "externalIdentifier": [
1162                        {
1163                          "identifier": "DEABCEXXXXXXXX12346",
1164                          "typeOfIdentifier": {
1165                            "value": "extendedG",
1166                            "extendedValueG": "evseId"
1167                          }
1168                        }
1169                      ],
1170                      "connectorType": {"value": "iec62196T2"},
1171                      "maxPowerAtSocket": 22000
1172                    },
1173                    {
```

Angabe plugAndCharge verpflichtend, falls diese Technik verfügbar

```
1174                    "externalIdentifier": [
1175                      {
1176                        "identifier": "DEABCEXXXXXXXX12347",
1177                        "typeOfIdentifier": {
1178                          "value": "extendedG",
1179                          "extendedValueG": "evseId"
1180                        }
1181                      }
1182                    ],
1183                    "connectorType": {"value": "chademo"},
1184                    "maxPowerAtSocket": 50000
1185                  }
1186                ],
1187                "electricEnergy": [
1188                  {
1189                    "energyProductName": {
1190                      "values": [
1191                        {
1192                          "lang": "de",
1193                          "value": "1a Strommix"
1194                        }
1195                      ]
1196                    },
1197                    "energyProductInformation": "https://_________________",
1198                    "renewableEnergyEvidence": "https://_________________",
1199                    "energyRate": [
1200                      {
1201                        "idG": "39E6F05D-070E-47EF-94D7-6DEDE1332B2B",
```

Eine (nicht näher spezifizierte) Energieform/Stromart, die per adHoc-Preis angeboten wird.
Electric Energy kann sowohl an einem ElectricChargingPoint definiert werden (wie hier) oder direkt an der E.I.Station

Hinweise siehe weiter oben Zeilen 774/775

Preisgestaltung für diese Stromart

Achtung: Eine AdHoc-Preisdefinition wird durch die EU-Regulierung gefordert, und insbesondere muss der AdHoc-Preis auch im dynamischen Datenteil angegeben (d.h. überschrieben) werden

Ad-Hoc Preis, zahlbar mit VISA oder über die Webseite

Eine we

```
1202          "ratePolicy": {"value": "adHoc"},
1203          "lastUpdated": "2025-01-10T11:13:51+01:00",
1204          "applicableCurrency": ["EUR"],
1205          "payment": {
1206            "paymentMeans": [
1207              {"value": "emv"},
1208              {"value": "website"}
1209            ],
1210            "brandsAccepted": [
1211              {"value": "visa"}
1212            ]
1213          },
1214          "energyPrice": [
1215            {
1216              "priceType": {"value": "pricePerKWh"},
1217              "value": 0.35
1218            }
1219          ]
1220        }
1221      ]
1222    },
1223    {
1224      "energyProductName": {
1225        "values": [
1226          {
1227            "lang": "de",
1228            "value": "Grüner Strom"
1229          }
```

Referer

```
1230                                    ]
1231                                  },
1232                                  "mobilityServiceProvider": [
1233                                    {
1234                                      "afacOrganisationByReference": {
1235                                        "organisationReference": {
1236                                          "targetClass": "afacIdentifiableOrganisation",
1237                                          "idG": "A1633959-F0BE-4F48-85CC-1A9FAE2BF34B"
1238                                        }
1239                                      }
1240                                    }
1241                                  ],
1242                                  "energyRate": [
1243                                    {
1244                                      "idG": "7DE39FF9-8F26-48FE-9981-07131534ACC2",
1245                                      "ratePolicy": {"value": "contract"},
1246                                      "lastUpdated": "2025-01-10T11:13:51+01:00",
1247                                      "applicableCurrency": ["EUR"],
1248                                      "rateName": {
1249                                        "values": [
1250                                          {
1251                                            "lang": "de",
1252                                            "value": "GrünTarif 1AB"
1253                                          }
1254                                        ]
1255                                      },
1256                                      "energyPrice": [
1257                                        {
```

```
1258                                    "priceType": {"value": "pricePerKWh"},
1259                                    "value": 0.3,
1260                                    "overallPeriod": {
1261                                      "overallStartTime": "2025-01-01T00:00:00+01:00",
1262                                      "overallEndTime": "2025-12-31T23:59:59+01:00",
1263                                      "validPeriod": [
1264                                        {
1265                                          "recurringTimePeriodOfDay": [
1266                                            {
1267                                              "startTimeOfPeriod": "21:00:00Z",
1268                                              "endTimeOfPeriod": "06:00:00Z"
1269                                            }
1270                                          ]
1271                                        }
1272                                      ]
1273                                    }
1274                                  },
1275                                  {
1276                                    "priceType": {"value": "pricePerKWh"},
1277                                    "value": 0.27,
1278                                    "overallPeriod": {
1279                                      "overallStartTime": "2025-01-01T00:00:00+01:00",
1280                                      "overallEndTime": "2025-12-31T23:59:59+01:00",
1281                                      "validPeriod": [
1282                                        {
1283                                          "recurringTimePeriodOfDay": [
1284                                            {
1285                                              "startTimeOfPeriod": "21:00:00Z",
```

Beispiel für einen komplexen Ta

```
1286                                                "endTimeOfPeriod": "06:00:00Z"
1287                                              }
1288                                            ]
1289                                          }
1290                                        ]
1291                                      }
1292                                    }
1293                                  ]
1294                                },
1295                                {
1296                                  "idG": "7F302DA6-CAA1-465E-A826-AE61C7CCACAF",
1297                                  "ratePolicy": {"value": "adHoc"},
1298                                  "lastUpdated": "2025-06-01T11:30:51+02:00",
1299                                  "applicableCurrency": ["EUR"],
1300                                  "rateName": {
1301                                    "values": [
1302                                      {
1303                                        "lang": "de",
1304                                        "value": "Komplexes Tarif-Beispiel"
1305                                      }
1306                                    ]
1307                                  },
1308                                  "energyPrice": [
1309                                    {
1310                                      "priceType": {"value": "pricePerKWh"},
1311                                      "value": 0.5
1312                                    },
1313                                    {
```

```
1314                                    "priceType": {"value": "pricePerMinute"},
1315                                    "value": 0.1,
1316                                    "priceCap": 12,
1317                                    "timeBasedApplicability": {"fromMinute": 240},
1318                                    "overallPeriod": {
1319                                      "overallStartTime": "2025-01-01T00:00:00+01:00",
1320                                      "exceptionPeriod": [
1321                                        {
1322                                          "recurringTimePeriodOfDay": [
1323                                            {
1324                                              "startTimeOfPeriod": "20:00:00Z",
1325                                              "endTimeOfPeriod": "08:00:00Z"
1326                                            }
1327                                          ]
1328                                        }
1329                                      ]
1330                                    }
1331                                  }
1332                                ]
1333                              }
1334                            ]
1335                          }
1336                        ]
1337                      }
1338                    },
1339                    {
1340                      "aegiElectricChargingPoint": {
1341                        "idG": "D8CF0A86-037F-449C-8BE2-5820EECC9036",
```

2. Ladepunkt

```
1342                    "versionG": "1",
1343                    "lastUpdated": "2025-01-10T11:13:51+01:00",
1344                    "deliveryUnit": {"value": "kWh"},
1345                    "currentType": {"value": "dc"},
1346                    "usageType": [
1347                      {"value": "electricVehicle"}
1348                    ],
1349                    "vehicleToGridCommunicationType": [
1350                      {"value": "iec619802"}
1351                    ],
1352                    "numberOfConnectors": 1,
1353                    "availableVoltage": [400],
1354                    "availableChargingPower": [40000],
1355                    "chargingMode": {"value": "mode4DC"},
1356                    "smartRechargingServices": [
1357                      {"value": "remoteMonitoring"},
1358                      {"value": "plugAndCharge"}
1359                    ],
1360                    "connector": [
1361                      {
1362                        "connectorType": {"value": "other"},
1363                        "otherConnector": "Some other connector type",
1364                        "maxPowerAtSocket": 40000
1365                      }
1366                    ]
1367                  }
1368                }
1369              ],
```

Angabe plugAndCharge verpflichtend, falls diese Technik verfügbar

```
1370                "electricEnergy": [
1371                  {
1372                    "energyRateByReference": [
1373                      {
1374                        "targetClass": "aegiEnergyRate",
1375                        "idG": "9C07640B-7ECE-46EC-A23F-417E99C53F33"
1376                      }
1377                    ]
1378                  }
1379                ],
1380                "supplementalFacility": []
1381              }
1382            ],
1383            "supplementalFacility": [
1384              {
1385                "afacSupplementalEquipment": {
1386                  "idG": "EF2D704B-AB23-45F2-BF1D-B08A2D919111",
1387                  "versionG": "1",
1388                  "availability": {"value": "available"},
1389                  "quantity": 2,
1390                  "regularlyCleaned": true,
1391                  "equipmentType": {"value": "toilet"}
1392                }
1393              },
1394              {
1395                "afacSupplementalServiceFacility": {
1396                  "idG": "SVC-001-CATERING",
1397                  "versionG": "1",
```

Definition auf Station-Ebene: Die gesamte 2. Ladestation nutzt den gleichen Tarif "GrünTarif 1AB" wie zuvor schon definiert bei einem Ladepunkt der ersten Ladestation.

Achtung: Jegliches Preisupdate im dynamischen Modellteil wirkt sich auf sämtliche Referenzierungen aus!

Definition auf Site-Ebene und damit gültig für alle enthaltenden Ladestationen: Diverse Ausstattungen.

! Die EU-Regulierung erwähnt insbesondere für Ladestationen die obligatorische Angabe von cateringService, bathroom und restingFacilities. !

2 regelmäßig gereinigte Toiletten

```
1398                        "serviceFacilityType": {"value": "cateringService"},
1399                        "availability": {"value": "notAvailable"}
1400                      }
1401                    },
1402                    {
1403                      "afacSupplementalEquipment": {
1404                        "idG": "EQP-002-BATHROOM",
1405                        "versionG": "1",
1406                        "equipmentType": {"value": "bathRoom"},
1407                        "availability": {"value": "notAvailable"}
1408                      }
1409                    },
1410                    {
1411                      "afacSupplementalEquipment": {
1412                        "idG": "EQP-001-RESTING",
1413                        "versionG": "1",
1414                        "nearby": true,
1415                        "equipmentType": {"value": "restingFacilities"},
1416                        "availability": {"value": "available"}
1417                      }
1418                    }
1419                  ]
1420                }
1421              ]
1422            }
1423          ]
1424        }
1425      }
```

Kein Catering Service, kein sonstiges Bad

Aufenthaltsraum in der Nähe

```
}
```